

Question_1:

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a **list of n numbers**. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A is **bigger than** B then this step's **output is 1**. B **removes** the number that was picked from their list.
2. If the number picked by A is **smaller than** B then this step's **output is 2**. A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is 0**. Both A and B **remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

$$1 \leq N \leq 10^6$$

$$1 \leq \text{num} \leq 10^9$$

Input format:

First line consists of a number n , size of the list provided.

Next line consists of n numbers separated by space.

Output format:

Output the required output list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 2 3

EXPECTED OUTPUT

2 2 0

Test Case 2

INPUT (STDIN)

5
5 7 1 2 3

EXPECTED OUTPUT

1 1 1 2 0 2 2 2

Code:

```
1  /*
2  Question_1:
3  A picks numbers from the start of the list and B picks numbers
4  from the end. If A>B output 1 and B removes, if A<B output 2
5  and A removes, and if A==B output 0 and both remove.
6  */
7
8  #include <iostream>
9  using namespace std;
10
11 int main()
12 {
13     int n;
14     cin >> n;
15
16     int a[n];
17     for(int i = 0; i < n; i++)
18     {
19         cin >> a[i];
20     }
21     int left = 0;
22     int right = n - 1;
23
24     while(left < n && right >= 0)
25     {
26         if(a[left] > a[right])
27         {
28             cout << 1 << " ";
29             right--;
30         }
31         else if(a[left] < a[right])
```

```
32         {
33             cout << 2 << " ";
34             left++;
35         }
36     else
37     {
38         cout << 0 << " ";
39         left++;
40         right--;
41     }
42 }
43 return 0;
44 }
```

Output:

Test Case_1:

```
3
1 2 3
2 2 0
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5
5 7 1 2 3
1 1 1 2 0 2 2 2
PS C:\Users\Prabin Yadav\
```